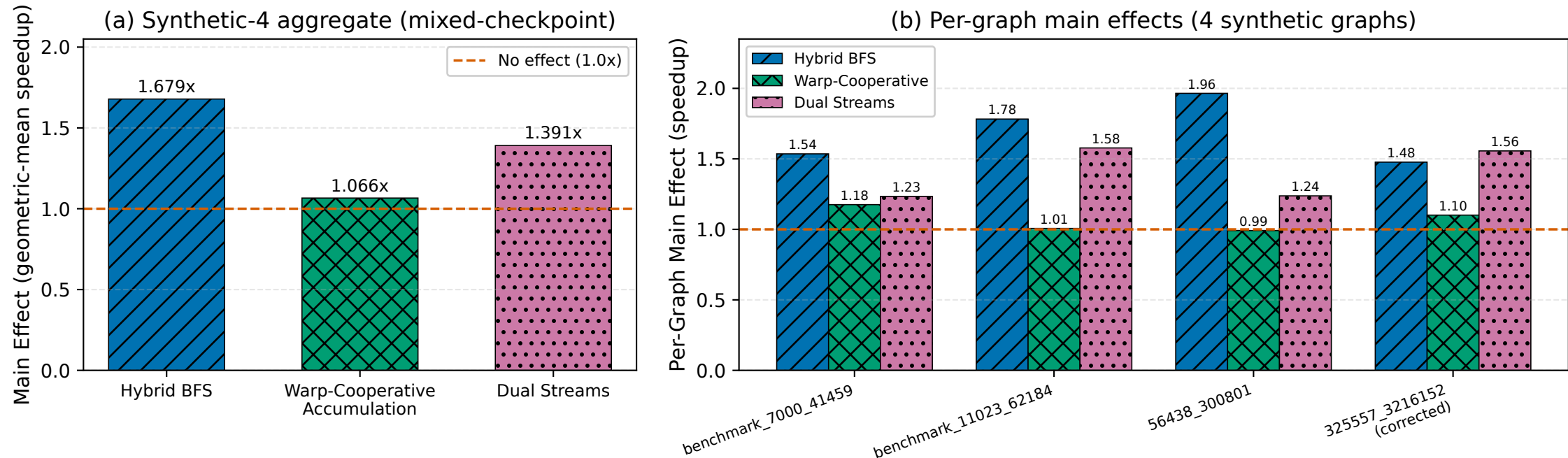


F4 Ablation Contributions (4 synthetic graphs; mixed-checkpoint aggregate)



Aggregate (a) is a MIXED-CHECKPOINT geometric mean: 3 graphs from job 2354994; corrected 325557 from job 2406254 (checkpoint 45352a3).
 Per-graph values (b) and the aggregate are distinct and not interchangeable. n=5 per configuration; the per-invocation untimed H1W1A1 warm-up is excluded from the 40 formal rows.
 Warp-Cooperative Accumulation is graph-dependent (56438_300801 < 1.0x). Not generalized to roadNet.